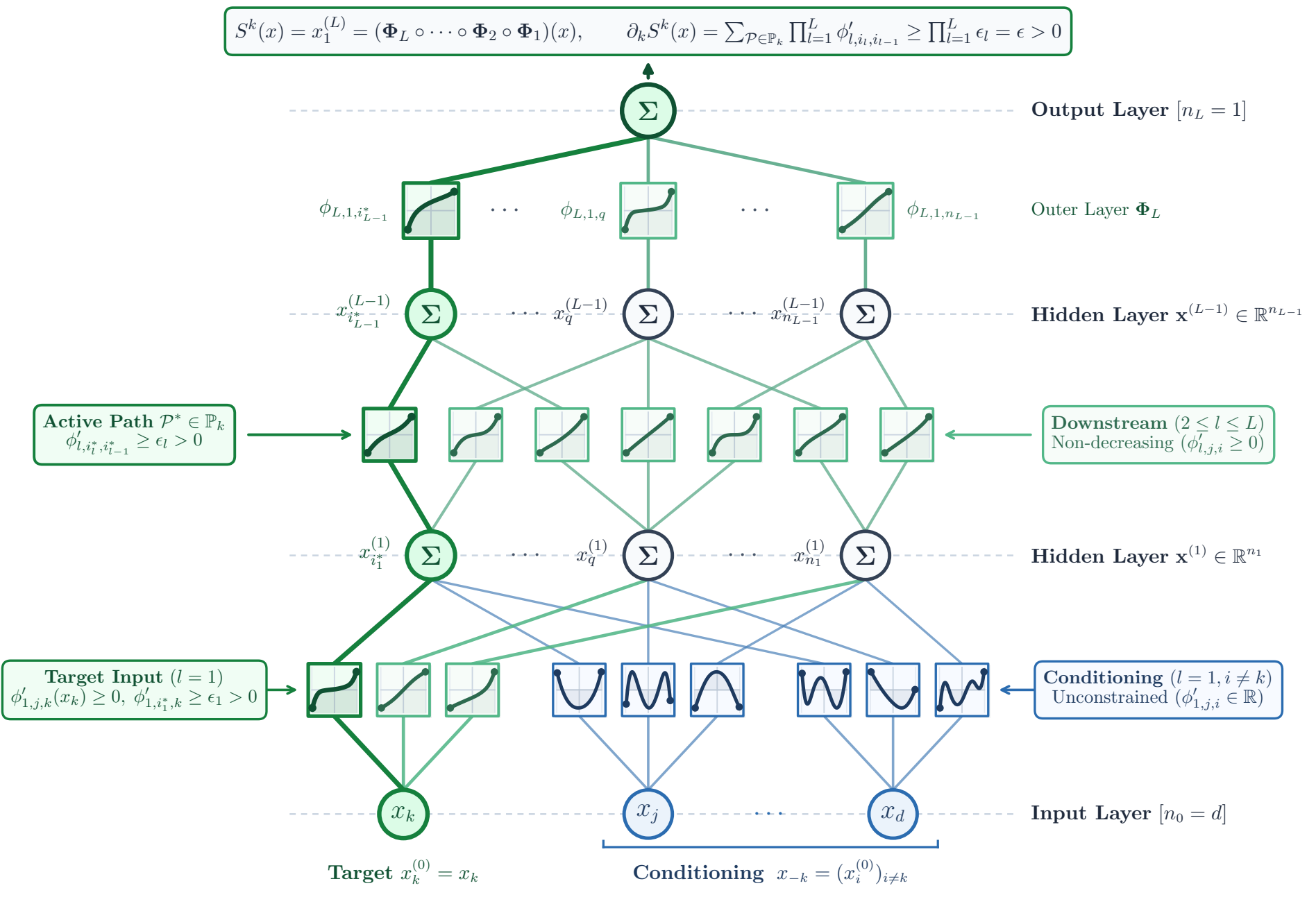

$S^k(x) = x_1^{(L)} = (\mathbf{\Phi}_L \circ \cdots \circ \mathbf{\Phi}_2 \circ \mathbf{\Phi}_1)(x), \qquad \partial_k S^k(x) = \sum_{\mathcal{P} \in \mathbb{P}_k} \prod_{l=1}^{L} \phi'_{l,i_l,i_{l-1}} \geq \prod_{l=1}^{L} \epsilon_l = \epsilon > 0$
Output Layer $[n_L = 1]$
$\phi_{L,1,i^*_{L-1}}$
$\phi_{L,1,q}$
$\phi_{L,1,n_{L-1}}$
Outer Layer $\mathbf{\Phi}_L$
$x^{(L-1)}_{i^*_{L-1}}$
$x^{(L-1)}_{q}$
$x^{(L-1)}_{n_{L-1}}$
Hidden Layer $\mathbf{x}^{(L-1)} \in \mathbb{R}^{n_{L-1}}$
Active Path $\mathcal{P}^* \in \mathbb{P}_k$
$\phi'_{l,i^*_l,i^*_{l-1}} \geq \epsilon_l > 0$
Downstream $(2 \leq l \leq L)$
Non-decreasing $(\phi'_{l,j,i} \geq 0)$
$x^{(1)}_{i^*_1}$
$x^{(1)}_{q}$
$x^{(1)}_{n_1}$
Hidden Layer $\mathbf{x}^{(1)} \in \mathbb{R}^{n_1}$
Target Input $(l = 1)$
$\phi'_{1,j,k}(x_k) \geq 0,\ \phi'_{1,i^*_1,k} \geq \epsilon_1 > 0$
Conditioning $(l = 1, i \neq k)$
Unconstrained $(\phi'_{1,j,i} \in \mathbb{R})$
x_k
x_j
x_d
Input Layer $[n_0 = d]$
Target $x^{(0)}_k = x_k$
Conditioning $x_{-k} = (x^{(0)}_i)_{i \neq k}$